

# Market Basket Analysis with Python – Snapdeal

## *Project Summary Report*

### Introduction

In this project, I analyzed customer behavior using the Snapdeal dataset. The dataset contains 800 customer responses. The analysis focused on purchasing behavior, customer satisfaction, customer segments, personalized recommendations, reviews, and transaction values.

### Key Findings

The customer data was cleaned before analysis by handling missing values, inconsistent entries, and unnecessary spaces in column names. After cleaning, the dataset contained 800 records with no missing values or duplicate rows.

Purchase behavior was fairly balanced across the different purchase-frequency groups. Clothing and Fashion was the most selected category, with 476 selections, followed by Beauty and Personal Care and Home and Kitchen. Browsing frequency was also fairly even across the four groups, with customer counts ranging from 195 to 204.

Shopping satisfaction was centered around the middle of the 1–5 scale. The average Shopping Satisfaction was 2.99, Personalized Recommendation Rating was 2.91, and Rating Accuracy was 2.96. A satisfaction rating of 3 was the most common response.

### Customer Segmentation

Customers were grouped using purchase frequency and shopping satisfaction. The main groups were Frequent Buyers, At-Risk Customers, Frequent But Not Satisfied, Infrequent But Satisfied, Occasional Shoppers, and Mixed/Moderate.

Frequent Buyers had the highest average transaction value at about 594,698, while Frequent But Not Satisfied customers had the lowest at about 523,176. These results show differences between the groups, but they do not prove that the segment itself causes higher or lower transaction values.

K-Means clustering was also used with review importance, recommendation rating, rating accuracy, and shopping satisfaction. Eight clusters were selected because they produced the highest tested silhouette score of 0.244. The score indicates some separation between the groups, but there is also considerable overlap, so the clusters are better viewed as general behavioral patterns.

### Recommendation and Review Insights

Customers who said recommendations were not helpful had an average Shopping Satisfaction of 2.89. The averages for customers who found recommendations helpful sometimes or yes were 3.08 and 3.02. This shows a small difference, but not a clear step-by-step relationship.

Recommendation engagement was also an area for improvement. There were 285 customers who reported no engagement with personalized recommendations, 272 who engaged sometimes, and 243 who engaged yes. Recommendation ratings from 1 to 3 accounted for 502 responses, compared with 298 ratings of 4 or 5.

Review reliability and helpfulness showed only small differences in satisfaction. Customers who relied heavily on reviews had an average satisfaction of 3.075, while customers who found reviews unhelpful had an average of 2.930.

## Transaction Behavior and Recommendations

The average transaction value was about 547,442, with a median of about 541,613. Customers with a satisfaction rating of 5 had an average transaction value of about 584,307, compared with about 518,181 for customers rating their satisfaction as 3. This suggests that higher satisfaction is associated with higher transaction values in this dataset, although the relationship is not perfectly consistent.

Based on the findings, the main recommendations are to improve the relevance and accuracy of personalized recommendations, use customer feedback to understand which recommendations are useful, and focus on improving the overall shopping experience for highly valuable customers.

## Conclusion

Overall, the analysis shows that customer behavior and satisfaction vary across the Snapdeal customer base. The main opportunities are to improve customer satisfaction, make personalized recommendations more relevant, and use customer behavior data to better understand different customer groups.

**Video Link:** [[https://drive.google.com/file/d/1QIsPNAIT3fdYNkY1Sbzcny\\_XWXjoLIao/view?usp=sharing](https://drive.google.com/file/d/1QIsPNAIT3fdYNkY1Sbzcny_XWXjoLIao/view?usp=sharing)]